

Relativistic corrections in satellite positioning systems



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1. Introduction

Satellite geolocation systems or GNSS (Global Navigation Satellite System) are introduced to determine the position of any observer on the Earth's surface and are based on a location principle known as trilateration, which consists of measuring different distances to points for which the location is already known.

If a constellation of satellites in orbit emits signals at a certain speed, any observer on the ground can know the distance that separates it from each satellite by measuring the time it takes to receive its signal. Thus, by the trilateration resulting from various distances, it could already know the specific point at which it is with respect to them, and by knowing the position of the satellites it would also know its position on the surface.

To do this, it needs a measurement of that time and in the case of total synchrony of times, it would be enough to send a single synchronous time with all the signals for the observer to compare the times received with his own to obtain those distances.

However, satellites move at high speeds and heights on the Earth's surface, so the relativistic effects become manifest and are also amplified by the speed of propagation of the signal, which is that of light in a vacuum.

Under such conditions, failure to introduce a correction for the measurement of eigentime on satellites would quickly lose synchrony after any synchronization and it would be impossible to establish such distances reliably and determine the position accurately from them.

It is important to note that relativistic correction is not the only one necessary in satellite geolocation because there are other effects that introduce errors in the measurement on the ground, such as the one derived from the rotation of the co-moving observer with the Earth and affected by its rotation (Sagnac effect) or the one derived from the chemical composition of the atmosphere that delays the signals.

This document is intended to detail the necessary relativistic corrections for height and speed. It also describes the basic principles of trilateration for satellite positioning and includes some analytical

mechanics demonstrations and notes that may be useful in the context of general relativity and satellite navigation.

All images included in this document are the author's creation or generated with Google's Nano Banana 2. For all the values that have been used in numerical approximations and other undeveloped expressions common to both special and general relativity, the following sources have been needed

Sources	
#1	Spacetime and Geometry: An Introduction to General Relativity , Carroll, S. M.
#2	Relativity in the Global Positioning System , Ashby N.
#3	Relativistic corrections in the European GNSS Galileo , ESA, Mudrak, A., De Simone, P., Lisi, M. (2015)
#4	Geodetic Reference System 1980 , IUGG
#5	Solar System Dynamics , JPL
#6	GPS Space Segment (ICD) , U.S. Department of Defense
#7	GLONASS Interface Control Document (ICD) , UNAVCO
#8	Galileo Interface Control Document (ICD) , European GNSS Service Centre
#9	BeiDou Interface Control Document (ICD) , China Satellite Navigation Office (CSNO)

2. Trilateration

2.1. Trilateration in the plane

Let us imagine a plane and a circle C_1 with all its points at a distance r_1 from its center o_1 . No observer on the circumference can determine its exact position by measuring the radius r_1 , only that it is on the circumference. Now imagine another circle with radius r_2 and center o_2 at an arbitrary distance of o_1 . Both circles have two points of intersection, p_1 and p_2 (figure 1).

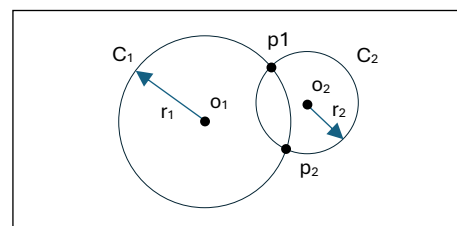


Figure 1

We can see that by knowing r_1 and r_2 the observer can determine that he is either in p_1 or in p_2 with respect to o_1 and o_2 , but not in which of them. We can also see in the very definition of the problem that there are infinite circles that pass through these two points, as many as we can draw with the center on the line that is defined by its centers because it is the mediatrix of the segment that joins p_1 and p_2 (figure 2).

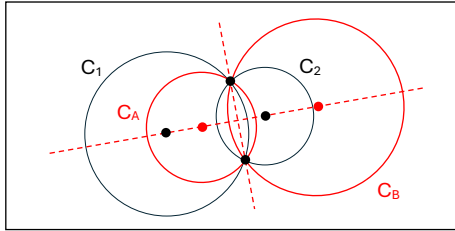


Figure 2

Let us now introduce a third circle C_3 with the only condition that it does not have the center on the line that joins the two previous centers (figure 3).

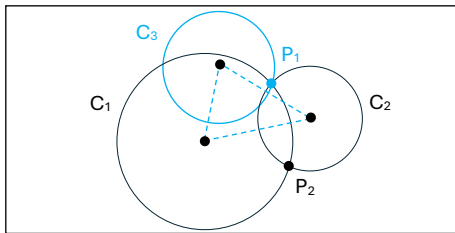


Figure 3

We see that there is no circumference that passing through one of the points of intersection of two circles can also pass through the other: the point p_1 is thus determined by the three radii of the circles C_1 , C_2 and C_3 . Determining a position in the plane using this procedure from three distances to points of known position is called trilateration.

2.2. Trilateration in space

Let us now try to extend this process to our three-dimensional space. Note that the equivalent of a circumference in the plane is the surface of a sphere S_1 where all its points are at a distance r_1 from its center o_1 . No observer on the surface of the sphere can determine its exact position, only that it is on that surface. Now imagine another sphere S_2

with radius r_2 and center o_2 at an arbitrary distance of o_1 . The surfaces of both spheres share an entire circumference as an intersection, C_{12} (figure 4).

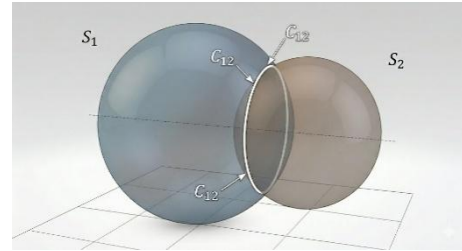


Figure 4

We find again, in the same definition of the problem, that there are infinite spheres whose surface shares the intersection C_{12} as long as their centers are on the line that joins both centers because the axis is perpendicular to the plane of the circumference intersection C_{12} that passes through their center (figure 5).

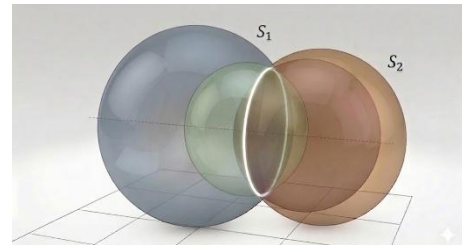


Figure 5

Let us now introduce a third sphere S_3 with the only condition that its center is not on the line that joins the centers of the previous two. Now the intersection of the surfaces of the three spheres is reduced to two points that are the intersection of the surface of S_3 with the circumference that results from the intersection of the other two C_{12} (figure 6).

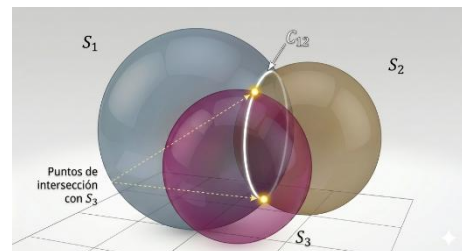


Figure 6

However, both points are equally possible, which makes a fourth sphere S_4 necessary. That fourth sphere cannot be on the same plane as the centers of the other three or we would be in a situation similar to that of the previous indeterminacies, finding infinite spheres whose surfaces have these two points as intersections. We see that it is impossible to find a fourth sphere whose center is not in that plane with a surface that passes at the same time through the two points intersection of the surfaces of the previous three, so that the point p_1 is already completely determined (figure 7).

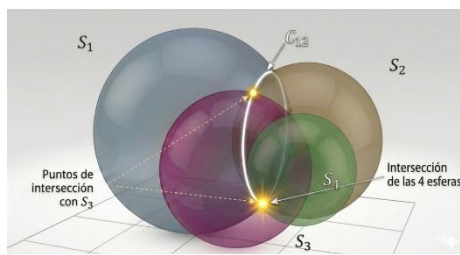


Figure 7

This is the positioning system in space that is used in GNSS systems.

2.3. Satellite Trilateration

Here it should be noted that when this procedure is used to determine satellite geolocation, the Earth itself resembles a sphere, so it would be sufficient to use three other spheres (three satellites) to determine the position. An observer on the Earth's surface would know at which of the two possible points at the intersection of the other three surfaces of the spheres he is because the other point of intersection would be inaccessible, so he would see his determined position (figure 8).

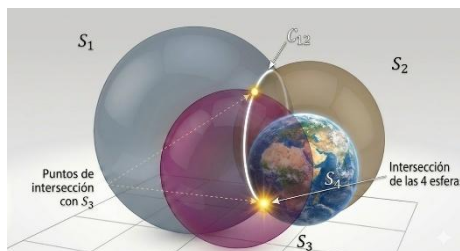


Figure 8

However, at least four satellites are used for reasons related to the ability to measure the observer's own time on the ground, which lacks the necessary precision and synchrony.

We know how to trilaterate a position in space from four fixed points, but how to solve trilateration when the centers of the spheres are mobile? This is the case of satellites, whose speed of movement is also remarkable and which are no longer even at the same point as the receiver on the surface when it receives its signals.

Thus, the satellites must send not only a temporary signal but also a position signal at the time of broadcast, so that the problem of geolocation, once other corrections have been resolved, translates into a problem of locating the points on the surface with respect to that of the satellites.

The satellites thus function as position references that self-determine based on the stability of their orbits and that they emit in the form of ephemeris, that is, orbital parameters that describe their trajectory at all times, and also need corrections for any alteration in their orbital mechanics. To ensure the accuracy of all measurements, both position and time, regular updates are sent to satellites from the ground.

For this reason, [non-geostationary Medium Earth orbits](#) with periods of less than one day are used, in order to be able to send frequent corrections from the ground using orbits in which the satellites are in free fall, as will be detailed later. As a curiosity, it is worth mentioning that the satellites also broadcast the constellation's almanac, that is, not only their own ephemeris but also that of the other satellites to help the observer on the ground to identify and check the signal of other satellites at all times.

3. Relativistic corrections

We are going to specify below the two relativistic corrections that must be implemented at the level of the satellites' own clock, given that of all the necessary corrections, these two are common to all observers on the surface of the globe regardless of where they are and hence they are the only ones that are configured in the

offset of the satellites' clock before they are launched.

The other corrections are not applied in the same way depending on the position and are therefore treated at the software level in the receiver. In the case of the necessary correction due to the rotation of the Earth (Sagnac) we observe that the velocity at the surface is not the same if it is measured at the equator (maximum) as if it is measured at the poles (null) and the same happens because of the delays in the signals induced by the composition of the atmosphere that is not homogeneous throughout the Earth.

It also happens that of all the necessary corrections, non-relativistic corrections (Sagnac, composition of the atmosphere, etc.) account for a minimal part of the total effect, although they cannot be neglected if we want to have precision in satellite navigation. The two relativistic effects represent almost the total effect where gravitational correction accounts for the largest part.

3.1. Imprecision in the temporal measurement of the observer on the ground

All signals travel at the speed of light in a vacuum (c), so microscopic errors in times translate into inadmissible errors when calculating distance. If we calculate the distances to the satellites with the formula $\sim 10^8 \text{ m/s}$

$$(3.1) \quad d = c \cdot \Delta t$$

We observe that every 1 of error (μs) translates into a distance error of hundreds of meters (m) and that considering only the distance to one of the satellites necessary to determine the position. That error would accumulate for relativistic purposes until it was corrected again from the ground, limiting the accuracy and usefulness of the entire system. $\mu\text{s} \sim 10^{-6} \text{ s} \sim 10^2 \text{ m}$

That is why we equip satellites with precise eigentime measurement technologies such as atomic clocks, but we cannot equip the receivers on the surface with the same technology. This is why we don't use just three satellites to determine the position on the surface, but four.

Using a fourth satellite allows the position to be calculated accurately without depending on the error in the measurement of the time of the observer on the ground, and also makes it possible to use it in devices such as portable navigators without them having precise technology to measure their own time, it also gives them the precision of measuring the time of the technology that equips the satellites. For practical reasons, even more satellites are used to prevent possible defects in some satellites or errors in the reception of their signals (such as the one caused by the bounce of signals between buildings in urban areas).

We see that ground observers calculate distances to satellites with the following equations, where x^i indicates the position of the satellite i for which the distance is to be calculated and the position of the observer of the form $d^i x$

$$(3.2) \quad (x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = d_i^2 = (c \cdot \Delta t_i)^2 = c^2 \Delta t_i^2$$

and therefore, calling the difference between the observer's own time and the satellites b assuming that all the satellites of the constellation share a total synchrony, we have to

$$(3.3) \quad (x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = d_i^2 = c^2 \cdot (t_i - t)^2 = c^2 b^2$$

and therefore

$$(3.4) \quad \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} - c \cdot b = 0$$

where we see that we can establish a similar equation for each satellite. We can also assume that the above equation is valid whether the measurement of proper time is delayed or advanced with respect to that of the constellation because the sign is absorbed in the square of the distance.

So we have an equation of four unknowns where one of them will be common and therefore four equations (four satellites) allow not only to define the position and that this unknown disappears but also to be determined and from it to adjust the receiver's own time, thus allowing any device to measure time with the precision of the atomic clocks carried by satellites.

3.2. Relativistic corrections for velocity

We know that the proper time of two observers in motion passes differently depending on their difference in velocities and we would therefore have to calculate the velocities of satellites and observers on the ground with respect to the same reference system, but as we pointed out before we cannot determine a uniform velocity on the surface.

Thus, we will limit ourselves to calculating the orbital velocity of the satellites according to the Earth's gravitational field using a reference system originating from its center that does not rotate with it, considering the observer on the surface almost static compared to the satellites in that reference system, and introducing the necessary software corrections at the receiver level. This system is called ECI (Earth-Centered Inertial). As we will also see numerically, this relationship between orbital velocities and those of surface observers is indeed high, which allows us to ensure that this system is valid for dealing with special relativity in the case of satellite navigation.

3.2.1. Free fall velocity (circular orbit)

For the field generated by the Earth we will consider the [Schwarzschild](#) metric, that is, we will consider the Earth a perfect radial symmetry, of the form

(3.5)

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2}$$

This solution imposes an asymptotically flat space-time that is already weak under the operating conditions of satellites where Newtonian mechanics could be perfectly applied.

For spherical symmetry and convenience we are going to calculate a fall restricted to the equatorial plane, i.e., . The modulus of velocity obtained with the calculations shall be applicable to all orbits of the satellites irrespective of their trajectory. $\theta = \pi/2 \Rightarrow \dot{\theta} = 0$

This simplification allows the spatial components of that velocity limited to the equatorial plane to be only two, the radial fall

and the linear (also contained in that plane and perpendicular to the radius and which develops in the angular coordinate). If we do not take advantage of the properties of symmetry, we would see two components appear in the plane perpendicular to the radius, one for each angular coordinate. $v_r \ v_l \phi$

This is further consistent with the [Christoffel symbols](#) of the metric under those conditions

(3.6)

$$\Gamma_{\theta\theta}^\phi = 0$$

$$\Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta = \cot \pi/2 = 0$$

Thus, returning to radial [geodesic](#)

(3.7)

$$\ddot{r} - \frac{r_s}{2r(r-r_s)} \dot{r}^2 - r \left(1 - \frac{r_s}{r}\right) (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 = 0$$

particularized for the equatorial plane ($\theta = \pi/2 \Rightarrow \dot{\theta} = 0$)

(3.8)

$$\ddot{r} - \frac{r_s}{2r(r-r_s)} \dot{r}^2 - r \left(1 - \frac{r_s}{r}\right) \dot{\phi}^2 + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 = 0$$

We can obtain the component of linear velocity by imposing and dividing between and therefore $\dot{r} = 0 \Rightarrow \dot{t}^2$

(3.9)

$$-r \left(1 - \frac{r_s}{r}\right) \dot{\phi}^2 + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 = 0$$

$$\Rightarrow -r \left(1 - \frac{r_s}{r}\right) \frac{v_l^2}{r^2} + c^2 \frac{r_s}{2r(r-r_s)} = 0$$

y What results

(3.10)

$$v_l^2 = \frac{GM}{(r-r_s)} \Rightarrow v_l = \frac{\sqrt{GM}}{\sqrt{r-r_s}}$$

which is the coordinate linear velocity in radial orbits. In the calculation of time dilation we must use the [Local speed](#) but in the case of satellite operation (i.e., for orbits) we have that this expression is valid because time dilation is not appreciable ($r \gg r_s \Rightarrow d\tau \approx dt$). We can further simplify the above

expression because and therefore use $(r - r_s) \approx r$

(3.11)

$$v_l = \frac{\sqrt{GM}}{r} = \sqrt{\frac{GM}{r}}$$

We will also observe that the calculations seem correct because for points far from the Schwarzschild radius the expression takes the Newtonian value of the velocity that results from equalizing the centripetal component of the radial field with its gravitational attraction towards the mass that generates it

(3.12)

$$F_g = F_c \Rightarrow G \frac{Mm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

3.2.2. Relativistic correction in free fall (circular orbit)

We are now in a position to calculate the time dilation experienced by static observers with respect to mobile observers in our reference system by recalling the Lorentz factor of time dilation according to the expression

(3.13)

$$\gamma = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

which can $v \ll c$ be approximated

as

(3.14)

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx \frac{1}{1 - \frac{1}{2} \frac{v^2}{c^2}}$$

Introducing this factor into the difference in the passage of proper time between static and moving observers, we have to

(3.15)

$$\begin{aligned} \Delta\tau_{\text{speed}} &= \frac{dt - d\tau}{dt} = 1 - \frac{d\tau}{dt} = 1 - \frac{1}{\gamma} \\ &= 1 - \left(1 - \frac{1}{2} \frac{v^2}{c^2}\right) = \frac{1}{2} \frac{v^2}{c^2} \\ &= \frac{GM}{2rc^2} \end{aligned}$$

which has a positive sign but indicates a delay in the time of the observers in motion over the static coordinates.

3.2.3. Relativistic speed correction applied to satellite geolocation

To obtain the difference between the time dilation in the satellites with respect to the observers on the ground we should use their difference in velocities, however we refer as the only valid difference to introduce in the offset of the clocks in the satellites the orbital velocity referring to an almost static surface of the Earth in comparison, ECI, due to lack of uniformity of velocities on the surface.

It is further found that the relationship between the velocities of ground observers and satellites follows a relationship of angular velocities and radii as follows

(3.16)

$$\begin{aligned} \frac{v_{\text{satellite}}}{v_{\text{land}}} &= \frac{\omega_{\text{satellite}}}{\omega_{\text{land}}} \frac{(r_t + h)}{r_t} \\ &= \frac{\omega_{\text{satellite}}}{\omega_{\text{land}}} \left(1 + \frac{h}{r_t}\right) \end{aligned}$$

Calling h the height of the satellites above the surface.

Since satellites have an [orbital period](#) of about 1/2 day we can assume that $\omega_{\text{satellite}} \approx 2 \times \omega_{\text{land}}$. In addition, h is several times (in the case of the Earth for geolocation orbits $h \gtrsim 3 \times r_t$) and therefore we have

(3.17)

$$\frac{v_{\text{satellite}}}{v_{\text{land}}} \gtrsim 2 \times (1 + 3) = 8$$

Returning to the previous expression, the difference in the passage of proper time between satellites and earth takes the form

(3.18)

$$\begin{aligned} \Delta\tau_{\text{speed}} &= \frac{1}{2c^2} (v_{\text{satellite}}^2 - v_{\text{land}}^2) \approx \\ &\frac{1}{2c^2} (v_{\text{satellite}}^2 - \frac{v_{\text{satellite}}^2}{64}) \end{aligned}$$

where we see that the second term is approximately 1.5% of the first, so we can disregard it with even greater certainty and establish that

(3.19)

$$\Delta\tau_{\text{speed}} = \Delta\tau_{\text{speed satellite}} - \Delta\tau_{\text{speed land}} \approx \Delta\tau_{\text{speed satellite}} = \frac{1}{2c^2} v_{\text{satellite}}^2 = \frac{GM}{2rc^2}$$

By making a spot check on the expression 3.11 for the orbital velocity with GRS80 values (#4)

(3.20)

$$V_{\text{land equator}} = \omega_{\text{land}} \times r_{\text{equator}} = 7.292.115 \times 10^{-11} \times 6.378.137 = 465,1 \text{ m/s}$$

We see that for the orbital velocity of satellites in a high orbit of 24 385 km, being higher in lower orbits, the error in disregarding it is

(3.21)

$$\varepsilon(\%) = \frac{(v_{\text{satellite}}^2 - v_{\text{land}}^2)}{v_{\text{satellite}}^2} \times 100 = \frac{(3.600^2 - 465,1^2)}{3.600^2} \times 100 = 1,67\%$$

Thus, the expression that will give us the phase difference that is introduced between the radio ground observers and the satellites in orbit at a height h due to their difference in speeds, introducing the height will be τ_t

(3.22)

$$\Delta\tau_{\text{speed}} = \frac{GM}{2(r_t + h)c^2}$$

Note that we do not use only h or height difference in the radial field because the orbital velocity of the satellite is obtained with respect to the origin of the reference system at the center of the Earth and that in the above expressions we can substitute the Schwarzschild radius to obtain

(3.23)

$$\Delta\tau_{\text{speed}} = \frac{GM}{2(r_t + h)c^2} = \frac{r_s}{4(r_t + h)}$$

3.3. Relativistic gravity corrections

We know that the proper time of two observers in a gravitational field passes differently depending on their potential difference, and we want to calculate those differences for Earth's gravitational field.

3.3.1. Relationship between the passage of time and the gravitational potential in spherical symmetries

We will use the same working metric ([Schwarzschild](#)) to determine the differences in gravitational potential expressed from the form

(3.24)

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2}$$

Static spherical symmetry imposes that the potential does not depend on the angular coordinate or the coordinate time, only on the radius, and it is important to note that we do not seek to define the space-time distance between two events with different radius values but the difference in the passage of proper time with respect to the coordinate in two different radius values.

Since g_{tt} already relates the proper time and the coordinate, we have that this relationship can be written as $d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2$

(3.25)

$$\frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

and since r and t are independent variables we have that r is constant when integrated in time and therefore

(3.26)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t$$

and that [it can be approached](#), in weak fields, such as

(3.27)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t \approx \left(1 - \frac{r_s}{2r}\right) \Delta t$$

which is the expression that will allow us to define the difference in the passage of proper time between two different radii with how $r_a \gamma r_b r_b > r_a$

(3.28)

$$\begin{aligned}\Delta\tau_{gravity} &= \Delta\tau_{grav r_b} - \Delta\tau_{grav r_a} \\ &\approx \left(1 - \frac{r_s}{2r_b}\right) - \left(1 - \frac{r_s}{2r_a}\right) \\ &= \frac{r_s}{2r_a} - \frac{r_s}{2r_b}\end{aligned}$$

that can be generalized to any generic r of the metric and an increment of the form Δr

(3.29)

$$\Delta\tau_{gravity} = \frac{r_s}{2r} - \frac{r_s}{2(r + \Delta r)}$$

3.3.2. Relativistic gravity correction applied to satellite geolocation

We can also rewrite the expression 3.29 with the classical form of the Newtonian potential by substituting the Schwarzschild radius and assigning the values for the radius of the Earth and that of the orbit of the satellites as $V =$

$$-\frac{GM}{r} r_t r_{sat}$$

(3.30)

$$\begin{aligned}\Delta\tau_{gravity} &= \frac{r_s}{2r_t} - \frac{r_s}{2r_{sat}} = \frac{GM}{r_t c^2} - \frac{GM}{r_{sat} c^2} \\ &= \frac{GM}{c^2} - \frac{GM}{r_{sat} c^2}\end{aligned}$$

Where is the value of the potential on the surface. This substitution is convenient because that value and the mean radius of the Earth are expressly included in the WRS80 model (#4).

3.3.3. Discrimination of near-mass effects

It seems convenient to check whether near-Earth masses influence the GNSS system by creating significant potential differences between the surface and satellites beyond those created by the Earth's field. Fortunately this is not the case and we are going to see that their influence is indeed negligible.

In the ECE in its general form

(3.31)

$$R_{ab} - \frac{1}{2} R g_{ab} = G_{ab} = \frac{8\pi G}{c^4} T_{ab}$$

all coupled equations have to be satisfied. For combined systems of n sources of gravity, all the equations of the form would have to be satisfied.

(3.32)

$$G_{ab}^{(n)} = R_{ab}^{(n)} - \frac{1}{2} R^{(n)} g_{ab}^{(n)} = \frac{8\pi G}{c^4} T_{ab}^{(n)}$$

which, as we can see, can never be generalized as a system of equations because the curvature tensors would belong to different metrics, so we cannot express the curvature as a linear expression of the one generated by the different masses.

Proving that we can discriminate the influence of other masses in the solar system thus becomes very difficult unless all the fields are weak and we can add potentials in the Newtonian way, that is, assuming the curved metric as the sum of a plane metric and a minimum perturbation that allows us to linearize it, in the form

(3.33)

$$g_{ab} = \eta_{ab} + x_{ab} \text{ with } x_{ab} \ll \eta_{ab}$$

where η_{ab} is the flat Minkowski metric, represents the metric we want to linearize, and considers the disturbance that adds curvature. In this way, the equations are linearized in the weak field regime.

The problem is simplified by verifying that all the masses of the solar system generate weak fields and that their influence on the Earth, which can already be added, is negligible on the Earth's field. We're going to check those fields for every significant mass in the solar system.

The influence of the Sun, by the principle of equivalence, will only manifest itself in tidal forces. Given that these tidal forces manifest themselves with a value inversely proportional to the cube of the distance and that the average of that distance is by definition 1 AU $\approx 150 \times 10^9$ m (#5, glossary) we can assume them negligible.

For the planets and satellites representative of the solar system, ordering their influence from highest to lowest on the Earth's gravitational field, being the potential of the Earth on its surface ($\Phi_{land} = -6,26 \times 10^7$ J/kg #4, Derived Physical Constants) and according to values in JPL (#5, Horizons, Astrodynamical Parameters), we can build the following table (table 1)

Body	D min (km)	Φ in D (J/kg)	Φ/Φ_{tierra}
Jupiter	5.88×108	-2.15×105	3.44×10-3
Saturn	1.20×109	-3.16×104	5.06×10-4
Moon	363.300	-1.35×104	2.16×10-4
Venus	3.82×107	-8.55×103	1.35×10-4
Uranus	2.57×109	-2.25×103	3.60×10-5
Neptune	4.30×109	-1.59×103	2.54×10-5
Mars	5.46×107	-7.84×102	1.25×10-5
Mercury	7.73×107	-2.85×102	4.56×10-6

Table 1

Where we can see that we can also disregard the potential differences they generate on the ground field for the purposes of the GNSS system.

As a curiosity, although we can disregard its gravitational effects in terms of calculating the time correction to be applied in satellites, we must take it into account in orbital calculations, and in fact in the GNSS systems in use today they are considered in the propagation of ephemeris.

3.4. Total relativistic correction

Since the eigentime for any observer in the radial field is advanced by its gravitational potential (the offset will have a positive sign) but is delayed by its velocity (offset with a negative sign), the total necessary correction derived from the relativistic effects for weak fields and circular orbits is

(3.34)

$$\Delta\tau_{\text{total}} = \Delta\tau_{\text{gravity}} - \Delta\tau_{\text{speed}} \\ \approx \left(\frac{r_s}{2r} - \frac{r_s}{2(r + \Delta r)} \right) - \frac{r_s}{4(r + \Delta r)}$$

which can be expressed as

(3.35)

$$\Delta\tau_{\text{total}} \approx \frac{r_s}{2r} - \frac{3r_s}{4(r + \Delta r)} \\ = \frac{2GM}{rc^2} - \frac{3GM}{2(r + \Delta r)c^2}$$

Thus, particularizing the expression for satellites with respect to surface observers, we have to

(3.36)

$\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{\text{sat}}c^2}$
--

which is the expression that will give us the offset to be corrected in satellites to avoid errors induced by relativistic effects.

3.5. Sensitivity analysis

We can perform a sensitivity analysis for all significant parameters and see how their variation influences the measurement of the passage of time in the case of two mobiles within a radial field and in particular the time calculations required in geolocation systems.

3.5.1. Sensitivity analysis for speed difference

As the orbit of satellites remains constant, we can assume that their orbital speed is also constant. In the same way we cannot expect changes in the rotational motion of the Earth and therefore the difference in velocities between satellites and ground observers will remain relatively constant, in addition to all the above considerations, ECI reference system and software correction.

It may be the case of observers on the surface moving at high speeds on the surface, but these corrections fall within the interferences to be analyzed in other software corrections that are outside the object of this document.

3.5.2. Sensitivity Analysis for Potential Difference

Recovering the previous expressions having defined $\tau ds^2 = -c^2 d\tau^2$ the proper tense such that and therefore

(3.37)

$$d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 \Rightarrow \frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

We can recover the concept of [compactness](#) and define the previous relationship as

(3.38)

$$\frac{d\tau}{dt} = \sqrt{(1 - u)}$$

which will give us the differential variation in the proper time as a function of the compactness of the radial field and which takes the form of the following graph (Figure 9)

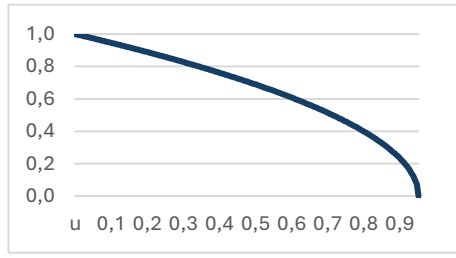


Figure 9

where we can see that at points far from the Schwarzschild radius where compactness tends to zero the time dilation is not appreciable, while for objects close to the Schwarzschild radius time tends to expand infinitely.

3.5.3. Application to the case of geolocation satellites

For the sensitivity analysis in satellite geolocation systems we find ourselves in a case where the field is weak (we can consider the compactness null) and the difference in relative velocities constant and therefore the following expression is completely valid

(3.39)

$$\Delta\tau_x = \left(\frac{r_s}{2(r_t + x)} - \frac{r_s}{2((r_t + x) + h)} \right)$$

and it will give us, by calling the height above the surface of the earth, the variation in the measure of time proper to the surface. x

Assuming that we move in the surface environment, we have to and therefore we can affirm that the variation in height does not alter the measurement in the positioning systems as long as we remain in the surface environment. $x \ll r_t r_t + x \approx r_t$

3.6. Verification of previous developments in operating systems

We want to validate previous developments by introducing concrete values for systems already in use on Earth. So for the GPS in particular we have that according to 3.36

$$\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{\text{sat}} c^2}$$

and with the values

$$(\#4) V_t = 6\,263\,686,085 \times 10 \text{ m}^2/\text{s}^2$$

$$(\#6) c = 299\,792\,458 \text{ m/s}$$

$$(\#4) GM = 3\,986\,005 \times 10^8 \text{ m}^3/\text{s}^2$$

$$(\#6) r_{\text{sat}} = 26,559,710 \text{ m (value of the semimajor axis of the true elliptical orbit)}$$

The formula to be applied is

$$\Delta\tau_{\text{total}} \approx$$

$$\begin{aligned} & \frac{6\,263\,686,0850 \times 10}{299\,792\,458^2} \\ & - \frac{3 \times 3\,986\,005 \times 10^8}{2 \times 26\,559\,710 \times 299\,792\,458^2} \\ & = 6,9693 \times 10^{-10} - 2,5047 \times 10^{-10} \end{aligned}$$

$$= 4,4646 \times 10^{-10}$$

So the satellites' own time is ahead of the Earth's surface for every second it spends on the surface. In other words, it will be brought forward daily $4,4646 \times 10^{-10} \text{ s}$

$$\Delta\tau_{\text{daily}} \approx$$

$$\begin{aligned} & (4,4646 \times 10^{-10} \times 3600 \times 24) \text{ s/day} \\ & = 38,574 \times 10^{-6} \text{ s/day} = 38,5 \mu\text{s/day} \end{aligned}$$

which is exactly the correction that is applied in the real GPS according to its ICD (#6, 3.3.1.1 Frequency Plan) where it is cited that being the nominal frequency of GPS clocks on the surface of 10.23 MHz that of satellites must be adjusted to

$$f' = (1 - \Delta\tau_{\text{total}}) \times f$$

$$= (1 - 4,4647 \times 10^{-10}) \times 10,23 \text{ MHz}$$

$$= 10,2299999954326 \text{ MHz}$$

This setting is introduced to GPS watches before launch. It has been proven that the expression returns correct values for other systems as well, although in Galileo the time correction is done via software and not by modifying the offset of the on-board clock. The results are shown in the following table (table 2)

GPS	
r_{sat}	26 559 710 (#6)
$\Delta\tau_{\text{total}}$ calculated	$4,4646 \times 10^{-10}$
$\Delta\tau_{\text{total}}$ in sources	$4,4647 \times 10^{-10}$ (#6)
$\Delta\tau_{\text{daily}}$	38.5 $\mu\text{s/day}$
GLONASS	
r_{sat}	25 478 136 (#7)
$\Delta\tau_{\text{total}}$ calculated	$4,3581 \times 10^{-10}$
$\Delta\tau_{\text{total}}$ in sources	$4.36 (\times 10^{-10})$ #7)
$\Delta\tau_{\text{daily}}$	37.6 $\mu\text{s/day}$

Galileo

r_{sat} 29 601 297 (#8)
 $\Delta\tau_{\text{total}}$ calculated 4,7216 $\times 10^{-10}$
 $\Delta\tau_{\text{total}}$ in sources 4.7219 ($\times 10^{-10}$)#3)
 $\Delta\tau_{\text{daily}}$ 40.7 $\mu\text{s/day}$

As a curiosity and proof of the previous conclusions about the sensitivity of the GPS system, the error introduced if we wanted to use it at 10 km altitude, which is the flight height of commercial aviation, would be

$$\begin{aligned}\Delta\tau_x &= \left(\frac{r_s}{2(r_t + x)} - \frac{r_s}{2((r_t + x) + h)} \right) \\ &= \frac{8,8701 \times 10^{-3}}{2} \left(\frac{1}{(6\,371 + 10) \times 10^3} \right. \\ &\quad \left. - \frac{1}{(6\,371 + 10 + 20\,200) \times 10^3} \right) \\ &= 5,2818 \times 10^{-10}\end{aligned}$$

which as we can see is equivalent to the calculation on the surface in a , that is, to the correction for relativity already predetermined in the GPS system. Also as a curiosity not to correct one's own time on satellites, the accumulated daily error in each measured distance with respect to GPS satellites would be $\frac{5,2818 \times 10^{-10}}{5,2922 \times 10^{-10}} = 99,80\%$

$$\begin{aligned}\Delta d_{\text{daily}} &\approx 38,5145 \times 10^{-6} \text{ s/day} \times c \\ &= 11\,546.35 \text{ m/day} = 11.5 \text{ km every day}\end{aligned}$$

4. Annexes and demonstrations

We are going to introduce some notes and general demonstrations of analytical mechanics and general relativity that may be useful in the context of this work, as well as others directly related to satellite navigation.

4.1. GNSS satellite orbits

GNSS satellites follow orbits in free fall with a low typical eccentricity () which means that we can assume perfect circles for the purposes of calculating linear velocity, i.e. the component tangential to the trajectory. This is not the case with regard to its ephemeris, where the calculation of position would be wrong if the exact value of the eccentricity was not implemented. $\lesssim 0,02$

Given that one of the fundamental design parameters is to cover as much of the surface as possible accurately, minimizing other costs such as the number of satellites

needed and that frequent updates have to be made from the ground, Medium Earth orbits are defined below about 35 000 km in altitude, around 20 000 km, avoiding the most immediate lower Van Allen belt that would destroy electronics and atomic clocks and which are the main reason why where we can have GNSS systems.

We know that linear [velocity](#) is $v = \sqrt{\frac{GM}{r}}$ and that we can also express as $v = \omega \times r$. We also know that it is the angular distance traveled per unit of time, so for a time T in which a complete circular path is traveled we have

$$(3.40)$$

$$v = \omega r \frac{2\pi}{T} r$$

and equating both expressions we have to

$$(3.41)$$

$$v = \sqrt{\frac{GM}{r}} = \omega \times r \frac{2\pi}{T} r \Rightarrow r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

which will give us the relationship between the radius of the stable circular orbit and a given period. And in the case of GNSS systems with a global reach, today we have (table 3)

System	Period (T)	r_{sat} (km)	h (km)
GPS (#6)	11h 58min	26.560	20.182
GLONASS (#7)	11h 15min	25.510	19.140
Galileo (#8)	14h 05min	29.593	23.222
BeiDou (#9)	12h 53min	27.906	21.535

Table 3

These orbits also meet other design conditions beyond the lack of friction necessary to allow stable circular orbits where angular momentum remains constant. If higher radios were used, higher signal strengths would also have to be implemented, and the cost of putting satellites into these orbits would also be higher.

4.2. Treatment of relativistic correction in technical publications (virial)

It is observed that the work that has been done in this document is a sufficient approximation to the published values for GNSS systems in use by drawing attention to the formula used in Ashby (#2, equation 35),

which is identical to the one used in this document substituting the radius for the semimajor axis of the elliptical and using properties of the WRS80 geoid, as follows

$$\Delta\tau = \frac{3GM_E}{2ac^2} + \frac{\Phi_0}{c^2}$$

the formulation developed in 3.36 being that we have changed signs and given the velocity correction a sign other than the gravitational one

$$\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{\text{sat}} c^2}$$

It is particularly remarkable that the correction found is the same when Ashby uses energy conservation in the Keplerian elliptical path, and here a perfect circular orbit has been assumed: Ashby uses the actual trajectory at all times and works with the eccentricity (called and found as the ratio between the distance from the center to the foci and the semimajor axis) even if it is minimal (in the GPS system the maximum eccentricity is a 0.02 but in Glonass, Galileo and BeiDou it is 0.000 and 0.001, with almost completely circular orbits [a #6](#), [#7](#), [#8](#) and [#9](#)).

This coincidence is produced by the virial theorem which specifies that a Keplerian elliptical orbit, when averaged in time, is identical in its mean values of kinetic and potential energy to a circular orbit whose radius is equal to the semi-major axis of the ellipse.

The analytical mechanics developed to consider eccentricity has been eliminated but any interested reader is recommended to work on its development, as well as on the Sagnac correction (much simpler), for a greater understanding of the effects of relativity on GNSS systems and its application not only at the satellite level, but as a receiver on the surface.

4.3. Some of Taylor's Approximations

In works of special and general relativity we will frequently find two ways for which it will be useful to use the following approximations of Taylor

$$f(x) = \sqrt{1-x} \approx 1 - \frac{1}{2}x \quad \text{Valid for } |x| \ll 1$$

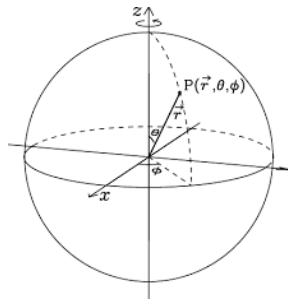
$$f(x) = (1+x)^n \approx 1 + nx \quad \text{Valid for } |x| \ll 1$$

In general relativity it appears as compactness and can be approximated to zero for weak fields (because it has an order of magnitude where n is the order of magnitude of the radius and b that of the mass) so the expression is perfectly valid. $u = \frac{r_s}{r} u \sim 10^{-27-n+b}$

In special relativity it appears as the relation $x \left(\frac{v}{c}\right)^2$ subtracting at the root of the denominator of the Lorentz transformation which makes the expression perfectly valid for velocities much lower than c .

4.4. Schwarzschild's solution for static spherical symmetries

We are going to demonstrate the Schwarzschild solution starting from the Minkowski metric in spherical coordinates of the form



$r \in [0, \infty)$
 $\theta \in [0, \pi]$
 $\phi \in [0, 2\pi)$

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$$

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

which we will frequently see written as follows to summarize the angular coordinates

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$$

And we will formulate a similar spherical symmetry where all free variables are expressed as functions of the radius and only of the radius (the expression for the angular coordinate remains unchanged to maintain symmetry) $f_i(r)$

$$ds^2 = -c^2 f_t(r) dt^2 + f_r(r) dr^2 + r^2 d\Omega^2$$

It will be convenient to write the functions of the previous expression as exponential functions multiplied by 2 to simplify all subsequent calculations and therefore $f_i(r)$

$$ds^2 = -c^2 e^{2f_t(r)} dt^2 + e^{2f_r(r)} dr^2 + r^2 d\Omega^2$$

We want to compute the functions, i.e., the terms $f_i(r)e^{f_i(r)}$ that make the spherical metric satisfy Einstein's field equation in a vacuum (and without cosmological cte Λ (because we are calculating the simple solution with an asymptotically flat spacetime) $T_{ab} = 0$)

$$R_{ab} - \frac{1}{2} R g_{ab} = \frac{8\pi G}{c^4} T_{ab} = 0$$

where we see that $R_{ab} = 0$ it is a necessary condition to satisfy the equation (if its scalar will also be null for any metric). Therefore, the problem is limited to finding the tensor for the postulated metric and knowing the value that the $R_{ab} = 0$ $R g_{ab} R_{ab}$ functions have $f_i(r)$ to satisfy in order for them to be satisfied. $R_{ab} = 0$

4.4.1. Calculating Christoffel Symbols

We will need to find all the Christoffel symbols of that metric necessary to calculate the components of the Ricci tensor to equal to zero and we will do it through the principle of least action.

To do this, we will define a Lagrangian of the form with an affine parameter in which L is held cte. By expressing the metric in tensor notation as and using the notation you can write the Lagrangian of the form $L = \left(\frac{ds}{d\lambda}\right)^2 \lambda \frac{\partial L}{\partial \lambda} = 0 ds^2 = g_{ab} dx^a dx^b \dot{x} = \frac{dx}{d\lambda} L = \left(\frac{ds}{d\lambda}\right)^2 = g_{ab} \dot{x}^a \dot{x}^b$

That Lagrangian applied to the metric exponentially that we had proposed is

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Where we have redeveloped the angular component to be able to find the Christoffel symbols associated with θ and ϕ . We will develop all the terms of the Euler-Lagrange expression of least-action with that Lagrangian ϕ

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

and we will compare the resulting terms with the geodesic equation in canonical form to obtain those of each coordinate Γ^a

$$\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

By developing those for our four-dimensional space for each coordinate according to the expressions of all the other possible pairs of coordinates and the metric the comparison of terms will be immediate. $\Gamma_{bc}^a x^a x^a x^b$

Development for the t-coordinate

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Remember that all variables are independent and their derivatives are independent (state space)

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) &= \frac{d}{d\lambda} (-2c^2 e^{2f_t(r)} \dot{t}) \\ &= \frac{d}{d\lambda} (-2c^2 e^{2f_t(r)}) \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} \end{aligned}$$

Applying the string rule that we can write as $=$ but not as $\frac{d}{d\lambda}$ because f is not derived with respect to the affine parameter, we have to $\frac{d}{d\lambda} [f(r(\lambda))] = \frac{df}{dr} \cdot \frac{dr}{d\lambda} \frac{d\lambda}{dr} \dot{r} f' \dot{r} \dot{r}$

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) &= [-4c^2 f'_t(r) e^{2f_t(r)} \dot{r}] \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} \\ \frac{\partial L}{\partial t} &= 0 \end{aligned}$$

Therefore the geodesic equation can be written as

$$[-4f'_t(r) c^2 e^{2f_t(r)} \dot{r}] \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} = 0$$

Dividing between we have to $-2c^2 e^{2f_t(r)}$

$$\ddot{t} + 2f'_t(r) \dot{r} \dot{t} = 0 \quad \text{compared to} \quad \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

It tells us that and the rest of Christoffel's symbols in t are null. $\Gamma_{rt}^t = \Gamma_{tr}^t = f'_t(r)$

Energy conservation

Here it is also important to note that with what the Lagrangian is conserved in for

any affine parameter and that from this conservation it also results that where $\frac{\partial L}{\partial t} = 0$ $t \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) = 0$ $\frac{\partial L}{\partial t} = -2c^2 e^{2f_t(r)} \dot{t}$

Therefore, it is a cte with respect to any related parameter. We will call this magnitude energy per unit of mass or specific energy, and its constancy conservation of energy. And for its general use we will eliminate all the ctes whose derivative is zero so as not to missize the equations in which we use it of the form $-2c^2 e^{2f_t(r)} \dot{t} \epsilon$

$$\epsilon = e^{2f_t(r)} \dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{t}} = \epsilon$$

This conservation is maintained in the event of [changes in coordinates](#) in y t , but not in changes in coordinates in r or ϕ

Development for the r-coordinate

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Remember that all variables are independent and their derivatives are independent (state space)

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{r}} \right) = \frac{d}{d\lambda} (2e^{2f_r(r)} \dot{r}) = \frac{d}{d\lambda} (2e^{2f_r(r)}) \dot{r} + 2e^{2f_r(r)} \ddot{r} =$$

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{r}} \right) = 4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r}$$

$$\frac{\partial L}{\partial r} = 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 + 2f'_r(r) e^{2f_r(r)} \dot{r}^2$$

Therefore the geodesic equation can be written and developed as

$$4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r} - [2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 + 2f'_r(r) e^{2f_r(r)} \dot{r}^2] = 0$$

$$4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r} - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 - 2f'_r(r) e^{2f_r(r)} \dot{r}^2 = 0$$

$$2e^{2f_r(r)} \ddot{r} + 4f'_r(r) e^{2f_r(r)} \dot{r}^2 - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 - 2f'_r(r) e^{2f_r(r)} \dot{r}^2 = 0$$

$$2e^{2f_r(r)} \ddot{r} + 2f'_r(r) e^{2f_r(r)} \dot{r}^2 - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 = 0$$

Dividing between we have to $2e^{2f_r(r)}$

$$\ddot{r} + f'_r(r) \dot{r}^2 - r(e^{-2f_r(r)})(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 f'_t(r) e^{2f_t(r)-2f_r(r)} \dot{t}^2 = 0$$

That compared with indicates that $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$

$$\Gamma_{rr}^r = f'_r(r)$$

$$\Gamma_{\theta\theta}^r = -r e^{-2f_r(r)}$$

$$\Gamma_{\phi\phi}^r = -r \sin^2 \theta e^{-2f_r(r)}$$

$$\Gamma_{tt}^r = c^2 f'_t(r) e^{2f_t(r)-2f_r(r)}$$

All other Christoffel symbols in r are null.

Development for the Coordinate θ

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Remember that all variables are independent and their derivatives are independent (state space)

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{d\lambda} (2r^2 \dot{\theta}) = \frac{d}{d\lambda} (2r^2) \dot{\theta} + 2r^2 \ddot{\theta} = 4r \dot{r} \dot{\theta} + 2r^2 \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = 2r^2 \sin \theta \cos \theta \dot{\phi}^2 = r^2 \sin 2\theta \dot{\phi}^2$$

Therefore the geodesic equation can be written as

$$4r \dot{r} \dot{\theta} + 2r^2 \ddot{\theta} - r^2 \sin 2\theta \dot{\phi}^2 = 0$$

Dividing between we have to $2r^2$

$$\ddot{\theta} + (2/r) \dot{r} \dot{\theta} - (1/2) \sin 2\theta \dot{\phi}^2 = 0$$

That compared with indicates that $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$

$$\Gamma_{r\theta}^{\theta} = \Gamma_{\theta r}^{\theta} = \frac{1}{r}$$

$$\Gamma_{\phi\phi}^{\theta} = -(1/2) \sin 2\theta$$

All other Christoffel symbols are null.

Development for the Coordinate ϕ

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Remember that all variables are independent and their derivatives are independent (state space)

$$\begin{aligned}\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\phi}}\right) &= \frac{d}{d\lambda}(2r^2 \sin^2 \theta \dot{\phi}) \\ &= \frac{d}{d\lambda}(2r^2 \sin^2 \theta) \dot{\phi} \\ &\quad + 2r^2 \sin^2 \theta \ddot{\phi}\end{aligned}$$

$$\begin{aligned}&= (4r^2 \sin \theta \cos \theta \dot{\theta} + 4r\dot{r} \sin^2 \theta) \dot{\phi} + \\ &2r^2 \sin^2 \theta \ddot{\phi} = 4r^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + \\ &4r\dot{r} \sin^2 \theta \dot{\phi} + 2r^2 \sin^2 \theta \ddot{\phi}\end{aligned}$$

$$\frac{\partial L}{\partial \phi} = 0$$

Therefore the geodesic equation can be written as

$$4r^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + 4r\dot{r} \sin^2 \theta \dot{\phi} + 2r^2 \sin^2 \theta \ddot{\phi} = 0$$

Dividing between we have to

$$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} + 2(1/r)\dot{r} \dot{\phi} = 0$$

That compared with indicates that $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$

$$\Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \cot \theta$$

$$\Gamma_{r\phi}^{\phi} = \Gamma_{\phi r}^{\phi} = \frac{1}{r}$$

All other Christoffel symbols are null.

ϕ

Conservation of angular momentum

Here it is also important to note that with what the Lagrangian is conserved in for any affine parameter and that from this conservation it also results that $\frac{\partial L}{\partial \phi} = 0$ where $\frac{\partial L}{\partial \phi} = 0 \Rightarrow \frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = 2r^2 \sin^2 \theta \ddot{\phi}$

Therefore, it is a cte with respect to any related parameter. We will call this magnitude angular momentum per unit mass or specific moment j (to distinguish it from the Lagrangian L) and its constancy conservation of angular momentum. And for its general use we will eliminate all the ctes whose derivative

is zero so as not to missize the equations in which we use it of the form $2r^2 \sin^2 \theta \ddot{\phi}$

$$j = r^2 \sin^2 \theta \dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{\phi}} = j$$

This conservation is maintained in the event of changes in coordinates in y, t , but not in changes in coordinates in r or ϕ

4.4.2. Calculating the components of the Ricci tensor R_{ab}

We are already in a position to obtain the Ricci components and equal them to zero to obtain the exponential functions of the proposed metric with the following summary table of non-null Christoffel symbols

Christoffel symbols in Schwarzschild		
$\Gamma_{rt}^t = \Gamma_{tr}^t = f_t'(r)$	$\Gamma_{\phi\phi}^r = -r \sin^2 \theta e^{-2f_r(r)}$	$\Gamma_{tt}^r = c^2 f_t'(r) e^{2f_t(r)-2f_r(r)}$
$\Gamma_{rr}^r = f_r'(r)$	$\Gamma_{\theta\theta}^r = -r e^{-2f_r(r)}$	$\Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \cot \theta$
$\Gamma_{r\theta}^{\theta} = \Gamma_{\theta r}^{\theta} = \frac{1}{r}$	$\Gamma_{\phi\phi}^{\theta} = -(1/2) \sin 2\theta$	$\Gamma_{r\phi}^{\phi} = \Gamma_{\phi r}^{\phi} = \frac{1}{r}$

We will develop only one of the tensor components to show the complexity of the calculation, and for this reason this will be the only explicit component of the calculus in this document. For the complete Schwarzschild solution we will directly introduce the values that can be obtained from multiple sources because this calculation is frequent in general relativity problems for the proposed metric. $R_{ab} R_{ab}$

Particular development for R_{tt}

$$R_{tt} = \partial_r \Gamma_{tt}^r + \Gamma_{tt}^r (\Gamma_{rr}^r + \Gamma_{r\theta}^{\theta} + \Gamma_{r\phi}^{\phi}) - \Gamma_{tr}^t \Gamma_{tt}^r$$

We have that the only partial derivative above corresponds to

$$\begin{aligned}\partial_r \Gamma_{tt}^r &= \partial_r (c^2 f_t'(r) e^{2f_t(r)-2f_r(r)}) \\ &= c^2 \left[f_t''(r) + 2(f_t'(r))^2 \right. \\ &\quad \left. - 2f_t'(r)f_r'(r) \right] e^{2f_t(r)-2f_r(r)}\end{aligned}$$

Replacing and developing we have to

$$\begin{aligned}
R_{tt} &= c^2 \left[f_t''(r) + 2(f_t'(r))^2 \right. \\
&\quad \left. - 2f_t'(r)f_r'(r) \right] e^{2f_t(r)-2f_r(r)} \\
&\quad + c^2 f_t'(r) e^{2f_t(r)-2f_r(r)} \left[\left(f_r'(r) \right) + \frac{1}{r} + \frac{1}{r} \right] \\
&\quad - c^2 f_t'(r)^2 e^{2f_t(r)-2f_r(r)} \\
R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + 2(f_t'(r))^2 \right. \\
&\quad \left. - 2f_t'(r)f_r'(r) \right. \\
&\quad \left. + f_t'(r) \left(\left(f_r'(r) \right) + \frac{1}{r} + \frac{1}{r} \right) \right. \\
&\quad \left. - f_t'(r)^2 \right] \\
R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + (f_t'(r))^2 \right. \\
&\quad \left. - 2f_t'(r)f_r'(r) \right. \\
&\quad \left. + f_t'(r) \left(\frac{rf_r'(r) + 2}{r} \right) \right] \\
R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 \right. \\
&\quad \left. - 2f_t'(r)f_r'(r) + f_t'(r)f_r'(r) \right. \\
&\quad \left. + \frac{2}{r}f_t'(r) \right] \\
R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 \right. \\
&\quad \left. - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right]
\end{aligned}$$

That added to the rest of the non-null components for the Ricci that we can extract from different sources ([#1](#), equations 5.14) we have

$$\begin{aligned}
R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 \right. \\
&\quad \left. - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right] \\
R_{rr} &= f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) \\
&\quad - \frac{2f_r'(r)}{r} \\
R_{\theta\theta} &= e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] - 1 \\
R_{\phi\phi} &= \sin^2 \theta \cdot R_{\theta\theta} \\
&= \sin^2 \theta (e^{-2f_r(r)} [1 \\
&\quad + r(f_t'(r) - f_r'(r))] - 1)
\end{aligned}$$

Where it is observed that all the crossed components disappear and that those that are not null depend on the radius and only on the radius, which is consistent with the initial assumption of spherical

symmetry with a diagonal metric dependent on the radius and only on the radius.

4.4.3. Integration into ECEs R_{ab}

4.4.3.1. R_{ab} Individual components of in the ECE

Now we will match all the non-null components to get the functions and searched $R_{ab} = 0$ $f_r(r) f_t(r)$

$$R_{tt} = 0 \Rightarrow c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right] = 0$$

$$R_{rr} = 0 \Rightarrow f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r} = 0$$

$$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] - 1 = 0$$

$$R_{\phi\phi} = 0 \Rightarrow \sin^2 \theta \cdot R_{\theta\theta} = \sin^2 \theta (e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] - 1) = 0$$

Of all the above coupled equations we will choose first because it is the one that implies immediate consequences to solve the system. $R_{\theta\theta} = 0$

$$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] = 1$$

$$\Rightarrow [1 + r(f_t'(r) - f_r'(r))] = e^{2f_r(r)}$$

$$\Rightarrow r(f_t'(r) - f_r'(r)) = e^{2f_r(r)} - 1$$

$$(R_{\theta\theta} = 0) \Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)} - 1}{r}$$

4.4.3.2. Component Comparison and $R_{tt}R_{rr}$

By showing all the components in a single expression we have

$$R_{tt} + R_{rr} + R_{\theta\theta} + R_{\phi\phi} = 0$$

or normalizing in the scalar

$$\begin{aligned}
\Rightarrow R &= g^{ab} R_{ab} = g^{tt} R_{tt} + g^{rr} R_{rr} + g^{\theta\theta} R_{\theta\theta} \\
&\quad + g^{\phi\phi} R_{\phi\phi} \\
&= \frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} + \frac{R_{\theta\theta}}{g_{\theta\theta}} + \frac{R_{\phi\phi}}{g_{\phi\phi}} \\
&= 0
\end{aligned}$$

Having proposed a spherical symmetry where they do not contain the functions we want to deduce, we will not be

able to use their components, but we can use those of t and r , of the form $g_{\theta\theta} \gamma g_{\phi\phi} f_t(r) \gamma f_r(r)$

$$\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = g^{tt}R_{tt} + g^{rr}R_{rr} = 0$$

and substituting the values of the individual expressions in this expression

$$\frac{c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right]}{-c^2 e^{2f_t(r)}} + \frac{R_{rr}f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} = 0$$

$$\frac{-e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right]}{e^{2f_t(r)}} + \frac{f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} = 0$$

$$-e^{-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right] - e^{2f_r(r)} \left[f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r} \right] = 0$$

and canceling term by term in $e^{-2f_r(r)}$

$$-e^{-2f_r(r)} \left[+\frac{2f_t'(r)}{r} \right] - e^{2f_r(r)} \left[-\frac{2f_r'(r)}{r} \right] = 0$$

and since the exponential can never be zero, we have to

$$\frac{2}{r} [f_t'(r) + f_r'(r)] = 0$$

and therefore

$$f_t'(r) + f_r'(r) = 0$$

$$\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0 \right) \Rightarrow f_t'(r) = -f_r'(r)$$

4.4.3.3. Integration of previous developments

Thus we have to apply the two expressions that we have concluded above we have to

$$(R_{\theta\theta} = 0) \Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)} - 1}{r} \quad \gamma$$

$$\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0 \right) \Rightarrow f_t'(r) = -f_r'(r)$$

$$\Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)} - 1}{r}$$

$$\Rightarrow 2r f_t'(r) = e^{-2f_t(r)} - 1$$

which is an ordinary equation of the first order with a solution

$$f_t(r) = \frac{1}{2} \ln \left(1 - \frac{C}{r} \right) \text{ where } C \text{ is an integration cte}$$

and according to our assumption we have to $f_t'(r) = -f_r'(r)$

$$f_r(r) = -\frac{1}{2} \ln \left(1 - \frac{C}{r} \right)$$

It is convenient to note since the above expressions can be written exponentially as

$$e^{2f_t(r)} = 1 - \frac{C}{r} \quad \gamma \quad e^{2f_r(r)} = \frac{1}{1 - \frac{C}{r}}$$

and therefore the initial metric can be written as

$$ds^2 = -c^2 e^{2f_t(r)} dt^2 + e^{2f_r(r)} dr^2 + r^2 d\Omega^2$$

$$ds^2 = -c^2 \left(1 - \frac{C}{r} \right) dt^2 + \left(\frac{1}{1 - \frac{C}{r}} \right) dr^2 + r^2 d\Omega^2$$

$$ds^2 = -c^2 \left(1 - \frac{C}{r} \right) dt^2 + \left(1 - \frac{C}{r} \right)^{-1} dr^2 + r^2 d\Omega^2$$

Where we have to determine the value of the integration cte C for the conditions of the problem.

4.4.4. Determination of mass (Newtonian limit)

To do this, we are going to make three assumptions based on classical mechanics.

The first is to assume that this integration cte C is directly proportional to the mass generated by the field, so that $C \propto M$

The second is the assumption that when r tends to infinity the metric takes the form of a flat space for that mass, so that

$$\lim_{r \rightarrow \infty} \left(1 - \frac{C \propto M}{r} \right) = 1, \quad \lim_{r \rightarrow \infty} \left(1 - \frac{C \propto M}{r} \right)^{-1} = 1$$

And the third is that in infinite r our equations have to behave as in classical

mechanics (Newtonian limit), that is, we will impose those conditions on the equations we have derived.

If we start from the geodesic equation for the radial component derived from the proposed particularized solution for r , according to its non-zero Christoffel symbols

$$\ddot{r} + \Gamma_{rr}^r \dot{r} \dot{r} + \Gamma_{\theta\theta}^r \dot{\theta} \dot{\theta} + \Gamma_{\phi\phi}^r \dot{\phi} \dot{\phi} + \Gamma_{tt}^r \dot{t} \dot{t} = 0 \quad \text{the related parameter being the proper tense } \tau$$

At the Newtonian limit, the speeds are much lower than those of light, so any component that has a spatial derivative is negligible compared to the other time-dependent components. On the other hand, when the affine parameter is eigentime, its derivative can be equal to 1 at infinity.

$$\Gamma_{rr}^r \dot{r} \dot{r} + \Gamma_{\theta\theta}^r \dot{\theta} \dot{\theta} + \Gamma_{\phi\phi}^r \dot{\phi} \dot{\phi} \approx 0 \text{ y } \dot{t} \approx 1$$

$$\ddot{r} + \Gamma_{tt}^r \dot{t} \dot{t} \approx 0 \Rightarrow \ddot{r} + \Gamma_{tt}^r \approx 0$$

Now we bring the expression of from the metric, Γ_{tt}^r

$$\Gamma_{tt}^r = \frac{1}{2} g^{rr} (\partial_t g_{rt} + \partial_t g_{tr} - \partial_r g_{tt})$$

Since we are working on a static solution, whose metric is not time-dependent, all partial derivatives with respect to time disappear

$$\Gamma_{tt}^r = \frac{1}{2} g^{rr} (-\partial_r g_{tt})$$

And since we have to $g^{rr} = 1/g_{rr}$

$$\Gamma_{tt}^r = -\frac{1}{2g_{rr}} (\partial_r g_{tt})$$

And we remember that

$$g_{tt} = -c^2 \left(1 - \frac{C}{r}\right) \text{ y } g_{rr} = \left(1 - \frac{C}{r}\right)^{-1}$$

$$\partial_r g_{tt} = \partial_r \left[-\left(1 - \frac{C}{r}\right) \right] = c^2 \frac{C}{r^2}$$

So

$$\begin{aligned} \Gamma_{tt}^r &= -\frac{1}{2g_{rr}} (\partial_r g_{tt}) = -\frac{1}{2g_{rr}} c^2 \frac{C}{r^2} \\ &= -c^2 \frac{C}{2r^2} \left(1 - \frac{C}{r}\right) \end{aligned}$$

Going back to the geodesic equation for radius,

$$\ddot{r} - \frac{C}{2r^2} c^2 \left(1 - \frac{C}{r}\right) \approx 0 \text{ and remembering the limit of the weak field we have to } \lim_{r \rightarrow \infty} \left(1 - \frac{C}{r}\right) = 1$$

$$\ddot{r} - \frac{C}{2r^2} [c^2] \approx 0 \Rightarrow \ddot{r} \approx \frac{C}{2r^2} [c^2]$$

Comparing with Newtonian acceleration, a , at the limit when r tends to infinity with what C will henceforth be called Schwarzschild's radius and which we will denote as $a = \frac{GM}{r^2} \ddot{r} = a \Rightarrow \frac{C}{2r^2} [c^2] = \frac{GM}{r^2} = \frac{2GM}{c^2} r_s = \frac{2GM}{c^2}$

And substituting in the metric,

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \text{ where } r_s = \frac{2GM}{c^2}$$

Which is the common way to display the Schwarzschild metric.

4.4.5. Useful evaluations about Schwarzschild radius

Working with the above expression to r_s find that, already introducing orders of magnitude for the fundamental ctes

$$r_s = \frac{2GM}{c^2} = \frac{2G}{c^2} M \sim \frac{10^{-11}}{(10^8)^2} M \sim \frac{10^{-11}}{10^{16}} \sim 10^{-27} M$$

And therefore by orders of magnitude

$$r_s \sim 10^{-27} M$$

We will call the relationship between r_s y M Schwarzschild radius per unit mass and denote it as u . We thus see that for any mass M , $r_m r_m \sim 10^{-27}$

We therefore find that the relation, which we will call compactness u , is a multiple of and therefore for orders of radius n the order of compactness will be times the order of the mass M , $\frac{r_s}{r} r_m 10^{-27-n}$

That is to say, calling u to compactness

$$u = \frac{r_s}{r} = M \frac{r_m}{r} \Rightarrow r \sim n, M \sim b \text{ y } r_m \sim 10^{-27} \Rightarrow u \sim 10^{-27-n+b}$$

Where we see that the order of the radius n and that of the mass M have to be significant in order to alter the compactness: only high masses and small radii will raise the

order of magnitude of the compactness and accepting a precision of order x we will have to

$x = b - n - 27 \Rightarrow b - n \geq x + 27$ where b is the order of mass and n is the order of radius

Or in another way, so that the curvature of space is significant as we will see below.

Being the expression of the metric

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Written now in terms of compactness

$$ds^2 = -c^2(1 - u)dt^2 + (1 - u)^{-1}dr^2 + r^2 d\Omega^2$$

We see that for any angular coordinate, at any radius of spherical symmetry, the order of compactness could cause it to be despised and therefore $\sim 10^{-27-n+b}$

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$$

Which exactly is the Minkowski metric.

The difference of orders of magnitude () proves to be a good criterion to disregard curvature and speak of weak fields where Newton's gravity, simpler, is completely valid. This difference is called the weak-field regime. $b - n$

As an example for the Earth imposing order 6 on the radius and 24 on the mass. $u \sim 10^{-9}$

4.4.6. Geodetic determination

We can now determine the geodesics in a radial gravitational field by retrieving the equations for the four dimensions and substituting the integration cte found

$$\ddot{t} + 2f'_t(r) \dot{r} \dot{t} = 0$$

$$\ddot{r} + f'_r(r) \dot{r}^2 - r \left(e^{-2f_r(r)} \right) (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 f'_t(r) e^{2f_t(r)-2f_r(r)} \dot{t}^2 = 0$$

$$\ddot{\theta} + (2/r) \dot{r} \dot{\theta} - (1/2) \sin 2\theta \dot{\phi}^2 = 0$$

$$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} + 2(1/r) \dot{r} \dot{\phi} = 0$$

And summarizing the functions and their derivatives

$$e^{2f_t(r)} = 1 - \frac{r_s}{r}$$

$$e^{2f_r(r)} = \frac{1}{1 - \frac{r_s}{r}}$$

$$f'_t(r) = \frac{r_s}{2r(r - r_s)}$$

$$f'_r(r) = \frac{-r_s}{2r(r - r_s)}$$

Thus we have that geodesics are

Geodesics in Schwarzschild	
t	$\ddot{t} + \frac{r_s}{r(r - r_s)} \dot{r} \dot{t} = 0$
r	$\ddot{r} - \frac{r_s}{2r(r - r_s)} \dot{r}^2 - r \left(1 - \frac{r_s}{r} \right) (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 \frac{r_s}{2r(r - r_s)} \dot{t}^2 = 0$
θ	$\ddot{\theta} + (2/r) \dot{r} \dot{\theta} - (1/2) \sin 2\theta \dot{\phi}^2 = 0$
ϕ	$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} + 2(1/r) \dot{r} \dot{\phi} = 0$

Also remembering that the following cyclical coordinates or conserved magnitudes are observed in the metric where the integration cte can already be replaced

Specific energy $\epsilon = e^{2f_t(r)} \dot{t} = \left(1 - \frac{r_s}{r}\right) \dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{t}} = \epsilon$

Angular momentum $j = r^2 \sin^2 \theta \dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Leftrightarrow \frac{\partial L}{\partial \dot{\phi}} = j$

4.4.7. Extended application of Schwarzschild's solution (Birkhoff's theorem)

Birkhoff's theorem states that every solution of ECE in a vacuum with spherical symmetry is necessarily static and asymptotically flat (its compactness u tends to 0 for $r \gg r_s$), so it coincides with Schwarzschild's metric.

Therefore, the Schwarzschild solution demonstrated above is still valid and equivalent to the one we would obtain by introducing time as a variable in the functions of the coordinates of the metric.

It also allows us to state that even if the radius in which the mass is concentrated varies over time, as long as the spherical symmetry is maintained, the solution will remain constant, or that considering that the

symmetry is maintained, the solution will not vary with changes in the internal density of, for example, the sun and the planets of the solar system including the Earth.

4.4.8. Local observers (shell)

Now let's observe that mobile phones never perceive a change of position with respect to themselves, they only see their own time pass. If they also follow a geodesic trajectory, they are in free fall in the radial field that generates it, so that by the principle of equivalence they only perceive a plane metric of the form

$$ds^2 = -c^2 d\tau^2$$

But reference systems that do not experience this free fall do perceive that movement, which also has to satisfy the invariance of for any observer, so that ds^2

$$ds_{cayendo}^2 = ds_{no\ cayendo}^2$$

that is,

$$-c^2 d\tau_{cayendo}^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt_{no\ cayendo}^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr_{nocayendo}^2 + r^2 d\Omega_{nocayendo}^2$$

which is the equation that allows us to deduce velocities close to the falling observer and which we call local observations or shell observations, because we consider that they occur on one of the spherical surfaces of r constant of the metric near the phenomenon we want to characterize.

These velocities are not the same as those that can be obtained by dividing the asymptotically flat metric by dt , which take velocity values valid for coordinate observers, i.e. at infinity, but which do not allow for impossible plane geometry addition operations with curvature.

In any case, the relationship between coordinate observer velocities and local or shell observers has to follow the time dilation relationship explained [here](#) and shown below

$$\frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

which, as we can see, is what allows us to use the time of the falling mobile to characterize the fall from another static mobile with respect to it in events far from the Schwarzschild radius.

4.5. Rigid coordinate changes applied to the principle of least action

We are going to reflect here some reflections on how changes in rigid coordinates affect the principle of least action.

For the Lagrangian and the metric expressed in tensor notation, an affine parameter is chosen in which L is held cte. $L = \left(\frac{ds}{d\lambda}\right)^2 ds^2 = g_{ab} dx^a dx^b \lambda \frac{\partial L}{\partial \lambda} = 0$

The notation is used with respect to this affine parameter and the variable change is defined $\dot{x} = \frac{dx}{d\lambda} x^a = x^a_0 + \alpha^a$ where it is a constant with respect to the affine parameter (condition 1) and the Lagrangian parameter (condition 2). α^a

It turns out that $\dot{x}^a = \dot{x}^a_0$ and because it disappears when derived (condition 1) and that we can continue to formulate the Euler-Lagrange equations with this Lagrangian (condition 2), although we will not be able to introduce it into generic tensor expressions because we are studying the case of a single variable change. $\ddot{x}^a = \ddot{x}^a_0 \alpha^a$

For each variable we have that the minimum action by Euler-Lagrange is of the form x^a

$$(Ec1) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

and we have to develop

$$T1 = , T2 = \text{and } T3 = \left(\frac{\partial L}{\partial \dot{x}^a} \right) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \left(\frac{\partial L}{\partial x^a} \right)$$

T1 Development

$$(T1) = \left(\frac{\partial L}{\partial \dot{x}^a} = \frac{\partial}{\partial \dot{x}^a} \left(\frac{ds}{d\lambda} \right)^2 \frac{\partial}{\partial \dot{x}^a} \frac{1}{d\lambda^2} g_{ab} dx^a dx^b \right)$$

Considering that all Lagrangian variables are independent, this expression will only take a non-zero value for and their cross-terms $\dot{x}^a$

$$\frac{\partial L}{\partial \dot{x}^a} = 2g_{ab} \dot{x}^b$$

Remembering that we cannot use the definition of the change of coordinates because it is only valid in the case that a is equal to b in the expression, so that $\dot{x}^a = \dot{x}^a_0$

$$\frac{\partial L}{\partial \dot{x}^a} = 2g_{ab} \dot{x}^b \neq 2g_{ab} \dot{x}^b_0$$

That is, a change of coordinates in will affect the term only in and its crossed terms (when b takes the value a in the previous expression). $x^a \frac{\partial L}{\partial \dot{x}^a} \dot{x}^a$

Examples

In Minkowski in spherical coordinates with the term $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ will only vary with changes in coordinates in r and but not in nor in t. $\theta\phi$

In Minkowski in Cartesian coordinates the term $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ will not vary with changes in coordinates in any variable.

In Schwarzschild with the term $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r, \theta)d\phi^2$ will vary with changes of coordinates in r but not in t, nor $\theta\phi$.

T2 Development

$$(T2) = \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \frac{d}{d\lambda} (2g_{ab} \dot{x}^b)$$

Considering that we can now find cross-terms where two variables depend on the parameter, we will have to perform the rule of the product of the form

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) &= \frac{d}{d\lambda} (2g_{ab} \dot{x}^b) = 2 \frac{d}{d\lambda} (g_{ab} \dot{x}^b) = \\ 2 \left(\frac{dg_{ab}}{d\lambda} \dot{x}^b + g_{ab} \frac{d\dot{x}^b}{d\lambda} \right) &= 2 \left(\frac{dg_{ab}}{d\lambda} \dot{x}^b + g_{ab} \ddot{x}^b \right) \\ &= 2 \frac{dg_{ab}}{d\lambda} \dot{x}^b + 2 g_{ab} \ddot{x}^b \end{aligned}$$

We decompose the derivative of the metric by entering a different index μ

$$\frac{dg_{ab}}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu} \frac{dx^\mu}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu} \dot{x}^\mu = (\partial_\mu g_{ab}) \dot{x}^\mu$$

Therefore, returning to the previous expression

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) &= 2 \frac{dg_{ab}}{d\lambda} \dot{x}^b + 2 g_{ab} \ddot{x}^b \\ &= 2 (\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b \\ &\quad + 2 g_{ab} \ddot{x}^b \end{aligned}$$

Remembering how the change of coordinates is defined, we have to analyze

how the previous expression is altered when the silent indices take the value a, at which time and $t\dot{x}^a = \dot{x}^a_0 \ddot{x}^a = \ddot{x}^a_0$

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) = 2 (\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b$$

The term will only be altered when a = b, where it will be sensitive to the change of coordinates in , so we can write that $g_{ab} \dot{x}^b x^a 2 g_{ab} \dot{x}^b \neq 2 g_{ab} \dot{x}^b_0$

The term will only be altered when a = b regardless of whether a = , since it will appear as a common factor in the sum of the partial derivatives of . Thus we can write that = and also observe that we do not violate any explicit rules in the syntax of tensor calculus (silent and free indices are respected). $(\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b \mu \dot{x}^a \mu (\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b 2 (\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b_0$

That is, a change of coordinates in will affect the term only in and its crossed terms (when b takes the value a in the previous expression), just as it happened to the term before making the total derivative with respect to the parameter. $x^a \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \dot{x}^a \frac{\partial L}{\partial \dot{x}^a}$

Examples

In Minkowski in spherical coordinates with the term $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ will only vary with changes in coordinates in r and but not in nor in t. $\theta\phi$

In Minkowski in Cartesian coordinates the term $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ will not vary with changes in coordinates in any variable.

In Schwarzschild with the term $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r, \theta)d\phi^2$ will vary with changes of coordinates in r but not in t, nor $\theta\phi$.

T3 Development

$$(T3) \frac{\partial L}{\partial x^a} = \frac{\partial}{\partial x^a} \left(\frac{ds}{d\lambda} \right)^2 = \left(\frac{\partial}{\partial x^a} \frac{1}{d\lambda^2} g_{ab} dx^a dx^b \right)$$

Here it is observed that considering all the independent variables, the expression will take a value other than zero in its terms crossed with the form x^a sólo para

$$\frac{\partial L}{\partial x^a} = \frac{\partial}{\partial x^a} \left(\frac{ds}{d\lambda} \right)^2 = \left(\frac{\partial}{\partial x^a} \frac{1}{d\lambda^2} g_{ab} dx^a dx^b \right) = (\partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu$$

That it will always be altered for any change of coordinates whenever it appears in a cross-term and will be transformed according to the definition $x^a x^a = x^a_0 + \alpha^a$ where it will appear explicitly because it is not eliminated in any derivation. α^a

If the Lagrangian is independent of the variable, its derivative in the variable will be null and therefore the Lagrangian will be cte in that derivative, i.e., if then and therefore $x^a \left(\frac{\partial L}{\partial x^a} \right) = 0 \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) = 0 \frac{\partial L}{\partial \dot{x}^a} = cte$. That will mean, which is the direct application of Noether's theorem, that we will find ourselves in the case of a conserved magnitude, . And therefore we can conclude that a change of coordinates in will not alter the conservation of the magnitude associated with that variable, since and therefore $\frac{\partial L}{\partial x^a} x^a \dot{x}^a = \dot{x}^a_0$

$$\frac{\partial L}{\partial x^a} = \frac{\partial L}{\partial x^a_0} = CTE$$

Examples

In Minkowski in spherical coordinates with the term $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ $\frac{\partial L}{\partial x^a}$ will only vary with changes in coordinates in r and but not in nor in t. The quantities conserved in t y $\theta \phi$ will continue to be conserved in the face of changes in coordinates themselves, but not in changes in r and θ .

In Minkowski in Cartesian coordinates the term $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ $\frac{\partial L}{\partial x^a}$ will not vary with changes in coordinates in any variable. The quantities conserved in the four variables will continue to be preserved in the event of any change in coordinates.

In Schwarzschild with the term $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r, \theta)d\phi^2$ $\frac{\partial L}{\partial x^a}$ will vary in the face of changes in coordinates in r but not in t, nor θ . The quantities conserved in t and $\theta \phi$ will continue to be conserved in the face of changes in coordinates themselves, but not in changes in r and θ .

Summary

We are going to integrate the above terms into the expression of the Ec1 geodesic, taking care not to cause duplication of indices that make them mute. Thus we have

$$(Ec1) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

That with $T_2 =$ and $T_3 =$ it is $\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \left(\frac{\partial L}{\partial x^a} \right)$

$$T_2 - T_3 = 0$$

We check before that

$$T_2 = 2 \left(\partial_\mu g_{ab} \right) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b$$

$$T_3 = \left(\partial_a g_{b\mu} \right) \dot{x}^b \dot{x}^\mu$$

But we have to alter the indexes in T_2 and T_3 to prevent them from violating the tensor syntax when we subtract T_2 and T_3 (the index b would appear three times in the summation).

$$\text{We can write like this } 2 \left(\partial_\mu g_{ab} \right) = \left(\partial_\mu g_{ab} + \partial_b g_{a\mu} \right)$$

$$\text{To rewrite } T_2 \text{ as } T_2 = \left(\partial_\mu g_{ab} + \partial_b g_{a\mu} \right) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b$$

So Ec1 results

$$\left(\partial_\mu g_{ab} + \partial_b g_{a\mu} \right) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b - \left(\partial_a g_{b\mu} \right) \dot{x}^b \dot{x}^\mu = 0$$

The canonical equation of the geodesic is according to the principle of least action for the coordinate x^a

$$(Ec2) \frac{d^2 x^a}{d\lambda^2} + \Gamma_{bc}^a \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

We want to integrate the above conclusions into Ec1 to derive how it is altered by the proposed variable change in and how its particular christoffels are altered. $x^a \Gamma_{bc}^a$

Altering the order of terms in the equal Ec1

$$2 g_{ab} \ddot{x}^b + \left(\partial_\mu g_{ab} + \partial_b g_{a\mu} \right) \dot{x}^\mu \dot{x}^b - \left(\partial_a g_{b\mu} \right) \dot{x}^b \dot{x}^\mu = 0$$

Drawing on the second term as a common factor $\dot{x}^b \dot{x}^\mu$

$$2 g_{ab} \ddot{x}^b + \left(\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu} \right) \dot{x}^b \dot{x}^\mu = 0$$

And seeking to resemble it to the canonical expression, it is important to bear in mind that

$$g^{ad} g_{ab} = \delta_b^d$$

So we're going to multiply the entire expression by a factor (1/2) and the inverse of the metric (we avoid introducing because we want to avoid doubling the index b and c we've already used it for a coordinate in the geodesic) resulting in $g^{ad} g_{ab}$

$$\delta_b^d \ddot{x}^b + (1/2)g^{ad} (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

And since the index b to d collapses, we have that and the expression is $\delta_b^d \ddot{x}^b \delta_b^d \ddot{x}^b = \ddot{x}^d$

$$\ddot{x}^d + (1/2)g^{ad} (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

And back to the comparison with the canonical equation

$$(Ec2) \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

We get the analytic expression for the Christoffels

$$(Ec3) \quad \Gamma_{b\mu}^d = (1/2)g^{ad} (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu})$$

So we see that in equation (Ec2) a rigid change of coordinates in the variable alters the expression of the metric in equation 3 (Ec3) and that of the Christoffels, and since the geodesic equation is covariant with changes in coordinates, the expressions of have to vary with the Christoffel to maintain equality. $x^a g^{ab} \Gamma_{bc}^a y \dot{x}^c$

4.6. Canonical geodesic equation according to the principle of least action

Here we are going to demonstrate the canonical form of the geodesic equation according to the principle of least action that is proposed as a good exercise to work with tensor notation, defining the Lagrangian

$$L = \frac{ds^2}{d\lambda^2} = \frac{g_{\mu\nu} dx^\mu dx^\nu}{d\lambda d\lambda} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad \text{with related notation and parameter } \frac{dx^a}{d\lambda} = \dot{x}^a \frac{dL}{d\lambda} = 0$$

Euler-Lagrange minimally acting formulation

$$\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0$$

Calculation of $\frac{\partial L}{\partial \dot{x}^\sigma}$

$$\begin{aligned} \frac{\partial L}{\partial \dot{x}^\sigma} &= \frac{\partial}{\partial \dot{x}^\sigma} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &= g_{\mu\nu} \dot{x}^\nu \frac{\partial}{\partial \dot{x}^\sigma} \dot{x}^\mu \\ &\quad + g_{\mu\nu} \dot{x}^\mu \frac{\partial}{\partial \dot{x}^\sigma} \dot{x}^\nu \end{aligned}$$

and remembering that in tensor notation $\frac{\partial \dot{x}^a}{\partial \dot{x}^b} = \delta_b^a$

$$\begin{aligned} \frac{\partial L}{\partial \dot{x}^\sigma} &= g_{\mu\nu} \dot{x}^\nu \delta_\sigma^\mu + g_{\mu\nu} \dot{x}^\mu \delta_\sigma^\nu \\ &= g_{\mu\nu} [\delta_\sigma^\mu \dot{x}^\nu + \delta_\sigma^\nu \dot{x}^\mu] \\ &= g_{\sigma\nu} \dot{x}^\nu + g_{\mu\sigma} \dot{x}^\mu = 2g_{\mu\sigma} \dot{x}^\mu \end{aligned}$$

Calculation of $\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma}$

$$\begin{aligned} \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} &= \frac{d}{d\lambda} 2g_{\sigma\mu} \dot{x}^\mu = 2 \frac{d}{d\lambda} g_{\sigma\mu} \dot{x}^\mu \\ &= 2\dot{x}^\mu \frac{d}{d\lambda} g_{\sigma\mu} + 2g_{\sigma\mu} \ddot{x}^\mu \end{aligned}$$

$\frac{d}{d\lambda} g_{\sigma\mu}$ it can be expressed as and therefore $\partial_a g_{\sigma\mu} \dot{x}^a$

$$\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} = 2\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + 2g_{\sigma\mu} \ddot{x}^\mu$$

Calculation of $\frac{\partial L}{\partial x^\sigma}$

$$\begin{aligned} \frac{\partial L}{\partial x^\sigma} &= \frac{\partial}{\partial x^\sigma} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = \dot{x}^\mu \dot{x}^\nu \frac{\partial}{\partial x^\sigma} g_{\mu\nu} \\ &= \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} \end{aligned}$$

Return to Euler-Lagrange

$$\begin{aligned} \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} &= 0 \Rightarrow \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0 \\ 2\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + 2g_{\sigma\mu} \ddot{x}^\mu - \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} &= 0 \end{aligned}$$

Dividing by 2

$$\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + g_{\sigma\mu} \ddot{x}^\mu - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} = 0$$

By symmetry one can write as and therefore

$$\partial_\nu g_{\sigma\mu} \dot{x}^\nu \dot{x}^\mu - \frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu$$

$$\frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu + g_{\sigma\mu} \ddot{x}^\mu - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} = 0$$

$$\frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} + g_{\sigma\mu} \ddot{x}^\mu = 0$$

$$\frac{1}{2} (\partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu - \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu}) + g_{\sigma\mu} \ddot{x}^\mu = 0$$

$$\frac{1}{2} \dot{x}^\mu \dot{x}^\nu (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) + g_{\sigma\mu} \ddot{x}^\mu = 0$$

Analytical formulation of Christoffel's symbols

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

Putting the Christoffels in Euler-Lagrange

$$\frac{1}{2} \dot{x}^\mu \dot{x}^\nu (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) + g_{\sigma\mu} \ddot{x}^\mu = 0$$

$$g_{\sigma\mu} \ddot{x}^\mu + g_{\sigma\lambda} \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$$

Multiplying by $g^{\rho\sigma}$

$$g^{\rho\sigma} g_{\sigma\mu} \ddot{x}^\mu + g^{\rho\sigma} g_{\sigma\lambda} \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$$

$$\delta_\mu^\rho \ddot{x}^\mu + \delta_\lambda^\rho \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$$

$$\ddot{x}^\rho + \Gamma_{\mu\nu}^\rho \dot{x}^\mu \dot{x}^\nu = 0$$

Result in canonical form

$$\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

5. Acknowledgements

This document arises as an end-of-course project for a remote training given by the National University of Distance Education (UNED) in its lifelong learning study program, "[Geometry of spacetime. Black holes and gravitational waves](#)", and hence contains general physics contents that exceed the context of satellite navigation.

Although the course has been aimed at explaining and applying general relativity in these two cosmological phenomena, I have decided to be humble in my aspirations regarding the syllabus and apply the physics exposed to a case where I feel more comfortable and can verify that the conclusions are indeed correct. So I have reserved cosmology to study it without fear of making mistakes that I leave for my leisure time.

This course has been my first university-level approach to general relativity and cosmology and for this I am extremely grateful to my professors Carlos Shahbazi Alonso and Pablo Bueno Gómez for their lessons, their initiative to inaugurate and facilitate this course and all the material they have made available to students. and to my classmates, for having been the most recommendable and fortunate company that could be expected in the endeavor.

If any error is found in the document, this will be the sole responsibility of the author, since as you have been able to read I have exceeded in demonstrations that I could have copied directly from more than one classic text or technical manual, but that I have developed in order to better understand the contents of the course.

And as a curiosity for those interested readers who are fans of science-fiction who probably know the film *Interstellar* (2014) by Christopher Nolan: at one point in the film the protagonists have to separate, some land on a planet near a black hole and others remain far from it and when they meet again time has passed much slower for those who have landed on the planet. They will find a graphic that explains it [here](#). I hope you savor this moment as much as I did when I developed this graph and thought that its consequences were exactly those shown in the film advised by none other than Kip Thorne, winner of the Nobel Prize in Physics in 2017 for his contribution to the understanding of gravitational waves.

And to point out some of those errors or any other topic of interest related to him, I leave here my contact: fer.urd@gmail.com

Fernando Urdiales
Madrid, May 2026